Nemani Anirudh

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Career Objective

Fourth-year ECE undergraduate currently interning as Sensors and Controls Intern at Seisou Labs Private Limited, Bengaluru, working on real-time embedded firmware, sensor integration, and thermal control systems. Skilled in Embedded C, MATLAB/Simulink, and communication protocols (I2C, SPI, UART) with hands-on experience deploying firmware on ESP32 and helped in designing custom made PCB, integrating industrial instrumentation, and supporting field commissioning. Seeking opportunities in embedded systems and hardware-software integration.

Education

Anurag University — Hyderabad, Telangana
B.Tech – Electronics and Communication Engineering | CGPA: 8.54 / 10 — *2023 – Present*

Trividyaa Junior College — Hyderabad, Telangana
Class XII – TSBIE | Percentage: 91% — *2021 – 2023*

The Secunderabad Public School — Hyderabad, Telangana
Class X – SSLC | Percentage: 77% — *2021*

Technical Skills

Microcontrollers & Processors: Arduino Uno/Nano, ESP32, 8086, 8051, TMS320C50 DSP Processor
Communication Protocols: I2C, SPI, UART, CAN (basics)
Programming Languages: Embedded C, Python (basics), Verilog HDL
Tools & Software: MATLAB, Simulink, Stateflow (basics), Xilinx Vivado, Mentor Graphics, Altium Designer (basics), KiCad (basics)
Hardware Skills: Circuit Design, PCB Design and Fabrication, Sensor Integration, Contactor-based Control
Sensors: ACS712 (Current), ZMPT101B (AC Voltage), HC-SR04 (Ultrasonic), Soil Moisture, DHT22, Pulse Rate Sensor, FSR
Industrial Instrumentation: RTD, Thermocouple, Temperature Transmitters, Process Control Instruments
Platforms: ESP32, Arduino, Xilinx Artix-7, Xilinx Zynq-7 Series
Domains: Embedded Firmware, IoT Systems, Signal Processing, FPGA Design, Machine Learning (Basics)
Soft Skills: Teamwork, Problem Solving, Leadership, Adaptability, Cross-functional Collaboration

Experience

Sensors and Controls Intern – Seisou Labs Private Limited — Bengaluru, Karnataka
Sensors, Controls, Instrumentation and Embedded Systems — *June 2026 – July 2026*

- Took end-to-end ownership of controls, instrumentation, and sensor integration for Seisou's cooling platform – covering firmware design, testing, and field troubleshooting on ESP32-based hardware.
- Developed and validated control logic for thermal comfort optimisation, condensation prevention, and water flow management; calibrated thermocouples, flow sensors, and SSR relay modules.
- Supported installation, commissioning, and field testing at client sites; maintained lab infrastructure and technical documentation; acquired PCB layout and board-level debugging skills.

Engineering Intern – Vedsri IT Solutions Pvt. Ltd. — Hyderabad, Telangana
Electronic Instrumentation and Industrial Temperature Sensing — *May 2025 – June 2025*

- Interned at an ISO 9001-certified process control instruments manufacturer; studied RTD and Thermocouple operating principles, accuracy, and signal behaviour under real industrial conditions.
- Assisted engineers in sensor selection for client applications by evaluating environment, temperature range, and accuracy requirements; analysed RTD vs. Thermocouple trade-offs across deployment scenarios.

Projects

AquaGuardAI - Smart Borewell Motor Protection System | *ESP32, Embedded C, I2C, UART, IoT* | GitHub — 2026 – 2027

- Engineered an ESP32-based adaptive protection system for 415V three-phase borewell motors using 3× ACS712 current sensors and ZMPT101B voltage sensor; implemented phase-wise RMS fault discrimination (phase loss, overload, over/under-voltage, dry-run) with trip-delay logic achieving fault detection within 100 ms–3.5 s per fault type.
- Integrated HC-SR04 ultrasonic sensor for real-time borewell water-level monitoring; implemented auto-restart after 30 s of stable conditions and safety lockout after 3 consecutive faults; status displayed on 16×2 I2C LCD with buzzer alerts and a WiFi cloud dashboard with offline hotspot fallback at 192.168.4.1.

Smart Assistive equipment for Children with Disabilities | *ESP32, Embedded C, I2C* | GitHub — Sep 2025 – Jan 2026

- Patented the concept and won **3rd prize** (30+ teams) at a department-wide hackathon; architected a wearable ESP32 system integrating Pulse Rate Sensor and FSR over I2C with haptic vibration feedback.
- Built a smart stylus with ML-based handwriting monitoring and incorporated personalised speech therapy and audio-video learning modules for children with ADHD and Autism.

FPGA-Based Real-Time Soil Nutrient Analyzer | *Xilinx Artix-7, Verilog RTL, I2C* | GitHub — Jan 2025 – Present

- Architected a synthesizable Verilog RTL pipeline on XC7A35T predicting 5 soil nutrients from 18-channel Vis-NIR data (AS7265x via I2C) within 847 µs at under 2 W.
- Built a Q16.16 fixed-point DSP pipeline with 5-tap Savitzky-Golay smoothing and DSP48E1 MAC-based regression – **14× faster** than equivalent ARM Cortex-M4 software.

ArithGen – Natural Language to Verilog RTL Generator | *Python, LLM, Verilog HDL* | GitHub — June 2025 – Present

- Built an end-to-end tool that converts plain English arithmetic descriptions into synthesizable Verilog RTL code targeting Xilinx FPGA deployment – from natural language input to simulation-ready module output.
- Implemented an LLM-based NL parsing pipeline to extract operand types, bit-widths, and arithmetic operations; generates valid Verilog modules from plain English prompts with no manual HDL writing required.

Certifications

MATLAB Onramp | *MathWorks* – MATLAB programming and data analysis fundamentals — Jan – Feb 2025

Simulink Onramp | *MathWorks* – Model-based design and Simulink simulation — Dec 2025 – Jan 2026

Python Basics | *Simplilearn* – Core Python programming and scripting — Jan – Feb 2026

Altium Designer Course | *Altium Education* – PCB design, schematic capture, and layout — July 2026

Workshops and Activities

PCB Design Workshop – Physitech Electronics — Anurag University, Hyderabad

Printed Circuit Board Design and Fabrication

- Fabricated a fully functional PCB – schematic entry, component placement, routing, and board-level testing.

Tech Lead – VLSI Club, Anurag University | **IETE Student Chapter**

- Led organisation of VLSI and FPGA workshops as VLSI Club Tech Lead; attended IETE seminars on embedded systems and communication technologies.

Personal Details

Gender: Male **Date of Birth:** 01 October 2005 **Languages Known:** Telugu, English, Hindi